

Friday the 13th and Traffic Accidents in Japan: A Natural Cultural Control Study of 1.9 Million Police Records

Running title: Friday the 13th: a Japan cultural-control study

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Abstract

Background: Evidence on whether Friday the 13th increases accident risk remains inconsistent and methodologically contested, with reported effects ranging from a 52% increase in hospital admissions to null findings. Japan, where 13 is not a traditionally salient unlucky number (the culturally emphasized inauspicious numbers are 4 and 9, not 13), provides a natural control population to test whether the “Friday the 13th effect” is culturally mediated.

Methods: We analyzed 1,884,793 police-recorded traffic accidents across Japan (2019-2024), comparing accident counts on 10 Friday the 13ths against other Fridays. We reproduced three established Western analytical approaches (Scanlon paired comparison, Näyhä negative binomial regression, and Lo multi-date ANOVA) and added three original analyses: (i) a case-crossover analysis on same-month Friday strata using three complementary tests (a month-stratified conditional Poisson regression, an exact sign test, and a sign-flip permutation test); (ii) a covariate-adjusted 47-prefecture negative binomial panel model with weather and holiday adjustment; and (iii) 47 prefecture-specific negative binomial fits summarized as a forest plot (Figure S3). Subgroup analyses examined accident severity, age groups, and time of day. Full statistical detail—including cluster-robust standard errors, small-cluster corrections, and residual diagnostics—is in Methods and the Supplementary Materials.

Results: Across all analytical approaches, no large Friday the 13th effect was detected. The Scanlon-method rate ratio was 1.04 (paired t-test $p = 0.47$), the Näyhä-method count ratio was 1.01 (95% confidence interval [CI], 0.54-1.91; $p = 0.97$), and the Lo-method ANOVA showed no between-group differences ($F = 0.16$, $p = 0.93$). The month-stratified conditional Poisson case-crossover regression gave a count ratio of 1.04 (95% CI, 0.95-1.14, sandwich- $t(G-1)$ $p = 0.40$); the companion exact sign test (8 of 10 pairs favored Fri13, $p = 0.11$) and sign-flip permutation test (exact $p = 0.45$ across all 1024 possible sign patterns; geometric count ratio 1.04) were also null. The 47-prefecture negative binomial panel model

with weather and holiday adjustment yielded a count ratio of 1.02 (95% CI, 0.97-1.08, two-way cluster $p = 0.36$); the confidence interval was centered near the null and excluded effects of the magnitude reported in early Western studies, although very small effects could not be ruled out. Prefecture-specific fits ranged from 0.59 (Tottori) to 1.29 (Iwate) with a median of 1.01; 5 of 47 nominally exceeded the null (binomial excess-test $p = 0.08$ versus $\alpha=0.05$), and after Bonferroni or Benjamini-Hochberg correction only Mie prefecture remained significant. Fatal accidents restricted to Friday-vs-Friday comparisons showed a crude rate ratio of 1.31 (Welch t-test raw $p = 0.09$, Bonferroni-adjusted $p = 0.70$), with a Poisson-approximation NB-adjusted count ratio of 1.23 (95% CI, 1.000-1.506, marginal lower bound) that did not survive multiple-comparison correction across the 8 pre-specified subgroups. All sensitivity analyses excluding COVID-period data, holidays, and Obon/New Year periods yielded consistent null results (RR range: 1.01-1.06).

Conclusions: In this large national dataset from a culture without a fear of the number 13, Friday the 13th was not associated with a detectable population-level increase in traffic accidents. Large effects of the magnitude reported by earlier Western studies can be excluded, but small effects cannot. Although the study was motivated by a cultural-mediation hypothesis, our null findings—combined with recent large studies from Western populations that also report no effect—are more consistent with a universal null effect than with a purely culturally mediated one. The absence of a large effect in Japan therefore adds to the growing evidence that Friday the 13th has, at most, only a very small influence on population-level accident rates in any country studied to date.

Keywords: Friday the 13th, triskaidekaphobia, traffic accidents, superstition, cultural epidemiology, Japan, cross-national comparison

Introduction

Is Friday the 13th truly unlucky? In 1993, Scanlon and colleagues published a now-classic study in the BMJ's Christmas issue reporting that hospital admissions from transport accidents in southwest England were 52% higher on Friday the 13th compared with Friday the 6th.¹ The finding captured public imagination and has been widely cited, but subsequent research across Western nations has produced conflicting results.

Näyhä reported a 63% higher risk of traffic death among Finnish women on Friday the 13th (adjusted rate ratio 1.63), though men showed no effect;² Radun and Summala directly contradicted this, finding no increase for either sex in Finnish injury accidents.³ Smith noted that Näyhä's analysis conflated date of death with date of accident—people who died on Friday the 13th may have been injured on a different day—potentially introducing survival bias.⁴ Dutch insurance data showed a 4% decrease.⁵ Subsequent larger studies have consistently found null results: Schuld et al. ($n = 27,914$ surgical patients, Germany),⁶ Lo et al. ($n = 49,094$ emergency department [ED] visits, USA; though penetrating trauma showed an isolated

odds ratio [OR] = 1.65),⁷ Ranganathan et al. (n = 19,747 surgical patients, Canada; adjusted OR = 1.02, 95% CI, 0.94-1.09),⁸ and Shekhar et al. (emergency medical services [EMS] volume, USA).⁹

A striking gap in this literature is the absence of data from cultures where 13 is not a traditionally salient unlucky number. All existing studies were conducted in Western nations with Christian cultural heritage, where triskaidekaphobia—fear of the number 13—is deeply embedded in social practices: buildings skip the 13th floor, airlines omit row 13, and hospitals avoid room 13.¹⁰ This shared cultural substrate makes it impossible to distinguish a genuine calendrical effect from a culturally mediated behavioral response.

Japan offers a compelling natural experiment. Japanese superstition centers on the numbers 4 (*shi*, homophonous with death) and 9 (*ku*, homophonous with suffering), not 13.¹¹ The asymmetry is concrete and observable: hotels and hospitals in Japan routinely skip floor 4 and room numbers containing 4, whereas the custom of omitting the 13th floor found in some Western contexts is not established in Japan.^{11,12} Elevator panels in Japanese buildings commonly omit 4F, and airlines skip row 4; no comparable convention exists for the number 13. While younger generations may be aware of the “Friday the 13th” horror film franchise, awareness of a foreign superstition is categorically different from holding a belief that alters behavior—a distinction that Phillips et al. demonstrated matters in their landmark study of mortality patterns among Chinese Americans on the 4th day of the month.¹² Crucially, the behavioral pathway requires not merely knowledge of a superstition but internalization sufficient to produce anxiety, avoidance, or altered risk-taking—a threshold that pop-cultural familiarity with a foreign film franchise is unlikely to reach.

Phillips et al.’s study provides the closest methodological precedent for our approach.¹² They found that Chinese and Japanese Americans, for whom the number 4 carries death-related associations, showed a peak in cardiac mortality on the 4th of the month (RR=1.13, 95% CI, 1.06-1.21), while White Americans showed no such pattern. Their design leveraged cultural variation within a single country to isolate belief-mediated effects. We extend this logic across countries: if the Friday the 13th effect is culturally mediated, it should be absent in Japan, where 13 carries no stigma.

We analyzed Japanese national police traffic accident records (2019-2024; approximately 1.9 million accidents) using the same analytical methods employed by previous Western studies, enabling direct cross-cultural comparison of effect estimates obtained through identical methodologies.

Methods

Data sources

Traffic accident data were obtained from the National Police Agency (NPA) of Japan open data portal, comprising individual accident records for 2019-2024. The dataset contains 1,884,793 records with date/time of occurrence, prefecture-level location, accident severity (fatal or injury), vehicle types involved, driver age, and time of day (daytime or nighttime). Records were aggregated to national daily counts (for the Scanlon, Näyhä, Lo, case-crossover, and subgroup analyses) and to 47 prefecture × daily counts (for

the panel-model covariate adjustment); no records were excluded, as the NPA dataset contains validated police records with no missing date fields. Weather data were obtained from the Japan Meteorological Agency (JMA) via daily cloud cover, precipitation, snowfall, and maximum snow depth series for 51 JMA prefectural stations (Hokkaido represented by 5 stations aggregated to the prefecture mean). Weather covariates were linked to each prefecture-day cell for the panel model; the national-level analyses used the corresponding national daily means. After daily aggregation, missing rates in the weather covariates were negligible (precipitation: 15 of 111,792 station-days = 0.013%; snowfall: 0; maximum snow depth: 1).

Study design

We identified 10 Friday the 13th dates during the study period (September 2019 through December 2024) and compared daily accident counts against control Fridays using multiple analytical frameworks. To maximize comparability with prior literature, we reproduced three established approaches alongside three original analyses (a case-crossover analysis, a covariate-adjusted 47-prefecture negative binomial panel model, and 47 prefecture-specific negative binomial fits summarized as a forest plot).

Scanlon method (paired comparison): Following Scanlon et al.,¹ we compared each Friday the 13th with the Friday the 6th of the same month using a paired t-test, with the Wilcoxon signed-rank test as a non-parametric robustness check. Normality of paired differences was confirmed by the Shapiro-Wilk test ($W=0.97$, $p = 0.91$).

Näyhä method (negative binomial regression): Following Näyhä,² we fitted a negative binomial regression model with Friday the 13th as a binary predictor, controlling for seasonal trends (harmonic terms for month), year, and cloud cover. The negative binomial model was selected over the Poisson model based on evidence of substantial overdispersion (Pearson chi-squared/df=20.8; Poisson AIC=9,660 vs. negative binomial AIC=4,953). As sex was not available in the NPA dataset, we performed age-stratified rather than sex-stratified analyses, using four groups: young (≤ 24 years), middle-lower (25-44), middle-upper (45-64), and elderly (≥ 65).

Lo method (multi-date ANOVA): Following Lo et al.,⁷ we compared accident counts across Fridays falling on the 6th, 13th, 20th, and 27th of each month using one-way ANOVA and the Kruskal-Wallis test, with post-hoc pairwise comparisons.

Case-crossover analysis (three tests, all date-level): Each Friday the 13th served as the case day, with the same-month other Fridays serving as controls (10 strata $\times$ 4 Fridays each = 40 case-control observations; every Fri13-containing month contained exactly one Fri13 and three other Fridays). Three complementary tests were run on the resulting pairs:

- *Month-stratified conditional Poisson regression (primary case-crossover reporting).* Following the case-crossover-as-conditional-Poisson framing,^{13,14} we eliminated the 10 stratum intercepts by conditioning on the stratum totals and maximized the resulting conditional likelihood in the Friday-the-13th log-count-ratio via Newton-Raphson. With $G = 10$ clusters, the Cameron-Miller¹⁵ finite-cluster correction factor $G/(G-1) = 1.111$ was applied to the cluster-by-stratum sandwich SE, and

inference was reported using both the $z(0.975) = 1.96$ and the $t(G-1) = 2.262$ critical values, with $t(G-1)$ as the primary reporting. Fisher information (iid) SE is retained in the machine-readable JSON for audit but is not reported in the manuscript, because Poisson equidispersion is violated at $\alpha \sim 0.02$ (see model diagnostics below) and would produce a misleadingly tight CI.

- *Exact sign test.* The two-sided binomial exact test against $p = 0.5$ was computed on the 10 case-control differences (ties excluded from the denominator).
- *Sign-flip paired log-ratio permutation test.* For each stratum we computed $\log(\text{fri13_count} / \text{mean}(\text{other Fridays}))$ and permuted the 10 signs; the 1024 possible sign patterns (2 to the tenth power) were fully enumerated for an exact p-value and cross-checked against 10,000 Monte-Carlo iterations (seed 20260723).¹⁶

Covariate-adjusted negative binomial panel model: We extended the Näyhä model to a 47-prefecture $\times$ daily panel with prefecture fixed effects, year/month/weekday fixed effects, and covariate adjustment for cloud cover, precipitation, snowfall, maximum snow depth, and three holiday indicators (national holiday, Obon, New Year). The variance was modeled as NB2, and the dispersion parameter α was estimated jointly by maximum likelihood ($\alpha_{\text{NB2}} \approx 0.023$ for the primary specification; NB1 parameterization gave $\alpha_{\text{NB1}} \approx 0.45$ as expected under the reparameterization identity $\text{var} = \mu(1+\alpha)$ vs. $\text{var} = \mu + \alpha\mu^2$). Standard errors were reported as two-way (prefecture, date) cluster-robust to accommodate day-level treatment (Bertrand-Duflo-Mullainathan concern for underestimated SEs under common-shock treatments). A 51-police-bureau panel (with the 5 Hokkaido bureaus retained separately) was fitted as a sensitivity check.

Prefecture-level heterogeneity (Figure S3). To display the between-prefecture distribution of the Fri13 effect underlying the panel-model estimand, we additionally fitted the same mean-model specification separately to each of the 47 prefecture subsets (weather + holiday adjustment retained; prefecture fixed effect dropped as degenerate within each single-prefecture subset). HC1 heteroscedasticity-robust standard errors were used because within a single-prefecture subset each date has exactly one observation, causing a nominal cluster-on-date to reduce to HC0 with a small finite-sample scaling; HC1 is the honest label. Weather covariates that were pairwise perfectly collinear within a prefecture ($|\text{corr}| \geq 0.999$; snowfall vs. maximum snow depth in 2 prefectures with 1-2 snow days) or zero-variance (6 prefectures with no snow) were dropped for those prefectures only. All 47 prefectures converged.

Model diagnostics. For the primary panel specification we pre-specified five diagnostic checks and one CRVE sensitivity check, all reported below and in the Supplementary Materials. Pre-specified diagnostics: (i) the MLE-estimated NB2 and NB1 dispersion parameters, judged robust if the $|\Delta|$ in the log-count ratio across parameterizations was below 0.01; (ii) Pearson residual summary statistics (mean, standard deviation, and tail proportions $|r| > 2$ and $|r| > 3$, benchmarked against the standard-normal expectation of 4.55% and 0.27%) and their visualization in Figure S1; (iii) a quasi-Poisson fit with Pearson χ^2/df scale factor as a companion overdispersion check; (iv) HC1 heteroscedasticity-robust standard errors alongside the primary two-way cluster SE; and (v) a seasonality-specification sensitivity comparing dummy month fixed effects, third-order harmonic terms ($\sin/\cos$ of $2\pi k \cdot \text{day}/365.25$ for $k = 1 \dots 3$), and a cubic B-spline of day-of-year. Because the 47-cluster count on the prefecture axis was less than the number of covariates

and the two-way cluster variance-covariance matrix was rank-deficient on that axis, we complemented the z-based cluster inference with a Cameron-Miller pair cluster bootstrap on the prefecture axis (500 iterations, seed 20260723, zero convergence failures). The bootstrap resamples only the prefecture axis, so its confidence interval under-covers the date-cluster variance and is reported as a prefecture-cluster misspecification sensitivity check rather than as an independent inference.

Subgroup and sensitivity analyses

Pre-specified subgroup analyses (Table 4) examined 8 subgroups selected before analysis: severity (fatal, injury) $\times$ 2, age (young $\leq$ 24, mid-lower 25-44, mid-upper 45-64, elderly $\geq$ 65) $\times$ 4, and time of day (daytime, nighttime) $\times$ 2. Each subgroup was fitted with the same national-level NB2 model (year, month, weekday, and holiday controls) for comparability across rows. Bonferroni correction was applied family-wise across the 8 subgroups (adjusted significance threshold: $\alpha = 0.05/8 = 0.00625$). Sensitivity analyses excluded: (1) the COVID-19 pandemic year (2020), (2) Obon and New Year holidays, (3) all national holidays, and (4) both the COVID-19 year and all national holidays simultaneously (i.e., scenario 1 combined with scenario 3).

Potential biases

As an ecological study using police-recorded data, potential biases include differential accident reporting across days and ascertainment bias from changes in policing patterns. These were partially addressed through the case-crossover design (same-month controls) and sensitivity analyses excluding holidays and the COVID-19 period, which represent the primary sources of systematic variation in reporting.

Power analysis

Because the number of Friday the 13ths was fixed by the calendar, we performed a sensitivity analysis of the minimum detectable effect rather than a post-hoc observed-effect power calculation (which would merely restate the p-value). With 10 Friday the 13ths and 303 other Fridays, the minimum detectable difference at 80% power and $\alpha = 0.05$ was 160.5 accidents per day (16.2% of the control mean). Effects of the magnitude reported by Scanlon (+52%) or Näyhä (+63%) would be detectable, but the observed effect (+3.9%) fell well below the detection threshold.

Ethical considerations

This study analyzed publicly available, anonymized accident-level records released by the National Police Agency of Japan through its open data portal. The records contain no personal identifiers, and all analyses were performed on daily aggregate counts derived from these records. Under the Japanese Ethical Guidelines for Medical and Biological Research Involving Human Subjects (2021 revision), research using publicly available data that do not include personal identifiers does not require ethics committee review (Article 3, Paragraph 1, Item 1). Accordingly, no institutional review board approval or waiver was sought. Because the study analyzes only aggregate, fully anonymized administrative records with no interaction with human participants, the Declaration of Helsinki is not directly applicable.

Statistical software

All analyses were performed using Python 3.14.3 with pandas 3.0.1, scipy 1.18.0, numpy 2.4.2, statsmodels 0.14.6, patsy 1.0.2, matplotlib 3.10.8, and pyarrow 25.0.0. Unless otherwise indicated, statistical tests were two-sided with a significance level of $\alpha = 0.05$; the sole exception was the one-sided binomial excess test used in the Prefecture-level heterogeneity analysis (Results), which asks specifically whether the observed count of prefectures with 95% CI excluding the null exceeded the null-hypothesis expectation. Full regression model coefficients, per-prefecture fit results, model-diagnostic JSON exports, and reproducibility code are available at <https://github.com/rehabilitation-collaboration/friday13th>.

Results

Descriptive statistics

During 2019-2024, Japan experienced 10 Friday the 13ths, with daily accident counts ranging from 713 (August 13, 2021, during Obon holiday) to 1,343 (December 13, 2019). The mean daily count on Friday the 13th was 1,026.9 (standard deviation [SD]=171.6), compared with 988.6 (SD=178.6) on other Fridays, yielding a crude rate ratio of 1.04 (Figure 1).

Reproduction of Western analytical methods

Scanlon method: All 10 Friday the 13th / Friday the 6th pairs from the same month were available for paired comparison. The mean difference was +44.0 accidents per day in favor of Friday the 13th (Friday 13th: 1,026.9 vs. Friday 6th: 982.9; rate ratio = 1.04). As with the Näyhä-method count ratio, this figure is a raw daily-count ratio rather than a formal incidence rate ratio (no exposure offset is applied). Neither the paired t-test ($t = 0.75$, $p = 0.47$) nor the Wilcoxon signed-rank test ($W = 20$, $p = 0.49$) reached significance. For comparison, Scanlon et al. reported 65 vs. 45 hospital admissions (raw admissions ratio of 1.44); after adjusting for a 1.4% reduction in traffic volume on Friday the 13th, they reported a headline risk ratio of 1.52 (a 52% increase; $p < 0.05$), from a far smaller sample.¹

Näyhä method: The negative binomial regression (AIC=4,953) yielded a count ratio of 1.01 (95% CI, 0.54-1.91, $p = 0.97$) for the Friday the 13th indicator. Because we modeled raw daily accident counts without a population exposure offset, this estimate is expressed as a count ratio rather than a true incidence rate ratio; the two are numerically equivalent when the underlying population at risk is effectively constant across Fridays within the study period. Age-stratified models showed uniformly null results: young count ratio=0.99, middle-lower=0.99, middle-upper=1.03, elderly=1.01 (all $p > 0.90$). For comparison, Näyhä reported an adjusted rate ratio of 1.63 for Finnish women and 1.02 for men.² Formal residual diagnostics for the panel version of this model (dispersion parameter, Pearson residuals, quasi-Poisson comparison, and rank-deficient cluster-robust variance estimator [CRVE] mitigation) are reported for the primary panel specification in the Model diagnostics paragraph below and in Supplementary Figure S1.

Lo method: One-way ANOVA across the four Friday groups (6th, 13th, 20th, 27th; $n=10$ each, verified from data) found no significant differences ($F=0.16$, $p = 0.93$; Kruskal-Wallis $H=0.35$, $p = 0.95$). Mean daily counts were 982.9, 1,026.9, 1,012.4, and 977.3, respectively. Post-hoc comparisons of the 13th against each control day were all non-significant (all $p > 0.50$). Lo et al. similarly found no overall increase in emergency department visits but reported an isolated increase in penetrating trauma (OR = 1.65, 95% CI, 1.04-2.61).⁷

Original analyses

Case-crossover (three date-level tests, Fri13 vs. same-month other Fridays): All 10 Fri13-containing months provided one Fri13 and three matched other Fridays ($n = 40$ case-control observations). The month-stratified conditional Poisson maximum-likelihood estimate gave a count ratio of 1.04, with the small-cluster Cameron-Miller correction yielding a sandwich- $t(G-1)$ 95% CI of 0.95-1.14 ($p = 0.40$); the corresponding sandwich- z inference gave a 95% CI of 0.96-1.12 ($p = 0.38$). The exact sign test on 10 same-month pairs gave 8 higher-than-control and 2 lower-than-control (ties = 0), for a two-sided binomial exact $p = 0.11$; individual case-to-control ratios ranged from 0.75 (August 2021, Obon period) to 1.20 (January 2023) and are shown as connected pairs in Supplementary Figure S2. The sign-flip paired log-ratio permutation test on the 1024 possible sign patterns (2 to the tenth power) gave an exact two-sided $p = 0.45$ (458 extreme patterns out of 1024; geometric count ratio 1.04); the Monte-Carlo estimate at 10,000 iterations agreed at $p = 0.45$ (MC SE = 0.005). All three case-crossover tests are consistent with the null. The arithmetic-mean summary statistic shown in Table 1 (mean ratio 1.05, $t = 1.05$, $p = 0.32$) is presented for reader transparency but is not the estimand used in the sign-flip permutation or conditional-Poisson tests reported here (which target the geometric mean; see Table 1 note).

Covariate-adjusted negative binomial panel model (primary): The 47-prefecture $\times$ daily panel model with weather, holiday, cloud-cover, precipitation, snowfall, and maximum-snow-depth adjustments, plus year/month/weekday fixed effects, yielded a Friday-the-13th count ratio of 1.02 (95% CI, 0.97-1.08, two-way (prefecture, date) cluster-robust $p = 0.36$; $n = 103,008$ prefecture-days). The panel model uses complete-case prefecture-days after excluding 16 cells with missing weather covariates (47 prefectures $\times$ 2,192 days = 103,024 total, minus 16 missing = 103,008). The 51-police-bureau sensitivity fit agreed to four decimals (count ratio 1.03, 95% CI, 0.97-1.08, $p = 0.35$). The Cameron-Miller pair cluster bootstrap on the prefecture axis gave a bootstrap 95% CI of 0.99-1.05 with two-sided $p = 0.18$; because it does not resample the date-clustered variance, its narrower CI is a prefecture-cluster misspecification sensitivity check, not a tighter primary estimate.

Model diagnostics (primary panel): The NB2 MLE dispersion parameter was $\alpha = 0.023$ (NB1 gave $\alpha = 0.45$; the log-count-ratio point estimate was invariant across parameterizations at $|\Delta| = 0.003$, well below the 0.01 spec-robustness threshold). Pearson residuals had mean +0.000 and SD 1.01, with 4.4% exceeding $|r| > 2$ (versus 4.55% expected under $N(0,1)$) and 0.69% exceeding $|r| > 3$ (versus 0.27% expected) — a mild fat tail consistent with a small number of extreme-event days that count regression cannot fully absorb (Figure S1). A quasi-Poisson fit gave the same point estimate to four decimals with a

Pearson χ^2/df scale factor of 1.44, comparable to the NB2 $\alpha = 0.023$ dispersion estimate. HC1 heteroscedasticity-robust SE (0.015) was about 56% of the two-way cluster SE (0.027), as expected under day-level common shocks. Replacing the month fixed effects with a third-order harmonic seasonality shifted the count ratio by ≤ 0.001 ; a cubic B-spline of day-of-year was rank-deficient against the intercept (B-spline partition-of-unity artifact) and is not reported.

Subgroup analyses (Table 4)

The 8 pre-specified subgroups (severity $\times 2$, age $\times 4$, time-of-day $\times 2$) all failed to reject the null at the family-wise Bonferroni threshold of $\alpha = 0.05/8 = 0.00625$. The fatal-accident row—which appears nominally significant at raw $p = 0.013$ under an alternative crude comparison against all non-Fri13 days (irrespective of weekday)—gave a raw two-sided Welch $p = 0.09$ (Bonferroni-adjusted $p = 0.70$) once restricted to the Friday-vs-Friday comparison used throughout this manuscript (mean 9.9 vs. 7.5 fatal accidents per Friday). The covariate-adjusted NB fit for the fatal subgroup did not converge (MLE optimizer failure at $\alpha \rightarrow 0$ on the sparse Fri13 fatal counts, mean $\sim 9.9/\text{day}$); we report the Poisson approximation as an approximate NB-adjusted count ratio of 1.23 (95% CI, 1.000-1.506; $p = 0.050$) and treat this row as approximate (see Table 4 footnote †). Nighttime accidents gave a crude rate ratio of 1.27 (Welch raw $p = 0.14$, NB-adjusted count ratio 1.07, $p = 0.25$); daytime gave 0.95 (Welch raw $p = 0.38$, NB-adjusted 0.96, $p = 0.37$). No age group showed a significant effect (all raw $p > 0.28$, all Bonferroni $p = 1.00$). None of the 8 subgroups survived Bonferroni correction, consistent with the null-everywhere pattern seen in the primary and case-crossover analyses.

Prefecture-level heterogeneity (Figure S3)

Fitting the primary weather-and-holiday-adjusted NB2 model separately to each of the 47 prefectures (all converged after handling of collinear snow covariates in Ehime and Okayama) produced Friday-the-13th count ratios ranging from 0.59 (95% CI, 0.32-1.09) in Tottori to 1.29 (95% CI, 0.99-1.67) in Iwate, with a median of 1.01. Five prefectures had 95% confidence intervals excluding 1.0 at the nominal $\alpha = 0.05$ level (11%, versus 2.25 expected under the null). The one-sided binomial test for an excess above the $\alpha = 0.05$ expectation gave $p = 0.08$, consistent with the null. Under a family-wise Bonferroni correction across the 47 prefectures ($\alpha_{\text{family}} = 0.05$, per-test threshold = 0.001) only Mie prefecture remained significant; the Benjamini-Hochberg false-discovery-rate procedure at $q = 0.05$ also retained only Mie. The 47-prefecture forest plot (Figure S3) shows this null-compatible spread centered on 1.0 rather than a systematic shift.

Sensitivity analyses

Results were robust across all exclusion scenarios (Table 2). Excluding COVID-19 data (2020) reduced the sample to 8 Friday the 13ths and yielded $\text{RR}=1.02$ ($p = 0.70$). Excluding Obon and New Year periods removed one Friday the 13th (August 2021) and produced $\text{RR}=1.06$ ($p = 0.26$). Excluding all holidays shifted the comparison toward the null ($\text{RR}=1.02$, $p = 0.67$). The most conservative analysis—excluding both COVID-19 data and holidays—produced $\text{RR}=1.01$ ($p = 0.90$), confirming the null finding.

Cross-national comparison

Table 3 and Figure 2 present a structured comparison of our results with prior studies. A consistent pattern emerges: the two earliest and smallest studies (Scanlon et al.,¹ $n = 110$; Näyhä,² $n = 41$ female deaths) reported significant effects, while all subsequent larger studies found null results regardless of the cultural setting. Importantly, our Japanese data—from a culture where 13 is not a traditionally salient unlucky number—produced effect estimates ($RR = 1.01-1.04$) that are virtually identical to those from recent Western studies (Radun and Summala,³ Schuld et al.,⁶ Ranganathan et al.⁸), suggesting that the early positive findings may have reflected small-sample variability rather than a genuine cultural effect.

Discussion

In the largest study to date examining Friday the 13th and traffic accidents, and the first conducted in a non-Western cultural context, we found no evidence that Friday the 13th increases accident risk in Japan. This null finding was consistent across five distinct analytical approaches, including three Western-derived methods (Scanlon, Näyhä, and Lo)—two of which (Scanlon and Näyhä) had reported overall positive findings in earlier Western populations, while the third (Lo) had reported an overall null result with an isolated positive finding for penetrating trauma only.

Interpretation in cultural context

Our results are consistent with two non-mutually exclusive interpretations. First, the “cultural mediation hypothesis” holds that superstition-related behavioral changes (e.g., heightened anxiety, altered driving patterns, avoidance behavior) occur only in populations that hold the relevant belief. Under this interpretation, Japan’s null result reflects the absence of triskaidekaphobia in Japanese culture, paralleling Phillips et al.’s finding that mortality peaks on the 4th of the month occurred only among Chinese and Japanese Americans for whom 4 symbolizes death.¹² Importantly, our design measures aggregate accident counts, not individual triskaidekaphobia, cognitions, or driving behavior; the cultural-mediation interpretation is therefore supportive rather than a direct test of belief-mediated behavior.

Second, the “null everywhere hypothesis” suggests that the original Western findings were artifacts of small samples and publication bias, and that Friday the 13th has no genuine effect in any population. This interpretation is supported by the pattern in Table 3: the only significant findings came from the two earliest and smallest studies, while all subsequent larger investigations—including those in Western countries with strong triskaidekaphobia—found null results. It is also worth noting that Phillips et al.’s methodological anchor (Ref [12])—although widely cited—has itself been subject to statistical critique regarding selective category inclusion,⁴ and later attempts to replicate belief-mediated mortality timing in Chinese populations outside the United States have reported mixed results; the cultural-mediation interpretation should therefore be viewed as a plausible framing rather than a confirmed mechanism.

On balance, the available evidence tilts toward the null-everywhere interpretation. The pattern in Table 3 is telling: significant results appeared only in the two smallest, earliest studies, while every subsequent

investigation with adequate sample size—regardless of cultural context—has found null results. Even if minor behavioral changes do occur among superstitious individuals (e.g., slightly more cautious driving, or avoidance of travel), these effects appear too small to detectably alter population-level accident rates in any country studied to date. Our power analysis indicates that we could detect effects of 16% or larger—well below Scanlon’s reported 52%—but not the 4% elevation we actually observed. Accordingly, our data exclude large effects but cannot rule out small ones.

The fatal accident signal

The fatal-accident row deserves comment because a crude comparison of Fri13 fatal counts against all non-Fri13 days (irrespective of weekday) yields a raw $p = 0.013$, which is nominally significant. Once the comparison is restricted to Friday-vs-Friday—the design used throughout this manuscript to hold day-of-week constant—the raw Welch two-sided p is 0.09, and the family-wise Bonferroni-adjusted p across the 8 pre-specified subgroups is 0.70 (Table 4). Because the covariate-adjusted NB2 maximum-likelihood estimator did not converge on the sparse Fri13 fatal counts (mean 9.9 accidents per Friday), the Poisson approximation shown in Table 4—count ratio 1.23 (95% CI, 1.0-1.5; exact bounds 0.9997-1.5061, i.e., the CI grazes rather than strictly excludes 1.0)—is reported as a bracket rather than a confirmatory estimate. The apparent elevation is consistent with Poisson fluctuation in small counts: with a mean of only 7.5 fatal accidents per non-Fri13 Friday, random variation alone produces day-to-day swings of 30% or more with appreciable probability. We report it for transparency but consider it most likely a chance finding—precisely the type of result that, if published in isolation, could perpetuate the Friday the 13th myth. Lo et al. similarly reported an isolated positive finding in one subgroup (penetrating trauma, OR = 1.65) while their overall analysis was null; that isolated finding is now generally interpreted as multiple-testing noise, and we consider the same interpretation appropriate here.⁷

Comparison with prior literature

Our systematic reproduction of Western analytical methods provides the strongest evidence to date that methodology is not the source of discrepant findings in this literature. When we applied Scanlon’s paired comparison to Japanese data with approximately 100 times more accidents per day than Scanlon’s original sample, the rate ratio was 1.04 versus Scanlon’s 1.44. When we applied Näyhä’s negative binomial model, our count ratio was 1.01 versus Näyhä’s 1.63 for Finnish women. The convergence of our null results across methods used in prior Western studies—two of which (Scanlon et al.¹ and Näyhä²) had reported positive overall findings—augmented by three complementary case-crossover tests (conditional Poisson $p = 0.40$, exact sign $p = 0.11$, sign-flip permutation $p = 0.45$) and a 47-prefecture forest plot centered on 1.01, together with the null results from recent large Western studies, strongly suggests that the early positive findings were statistical flukes rather than genuine effects.

Strengths and limitations

This study has several strengths: it is the first from a non-Western cultural context, it employs the largest sample in this literature (1.9 million accidents), it systematically reproduces prior methods for direct comparison, and it reports a comprehensive suite of sensitivity analyses.

Several limitations should be acknowledged. First, with only 10 Friday the 13ths, our power to detect small effects was limited (minimum detectable effect: 16.2%); effects smaller than approximately 10% cannot be excluded by our data, though such small effects would lack practical significance for public health or traffic safety policy. The effects reported by Scanlon (+52%) and Näyhä (+63%) would have been easily detectable. Second, cultural exposure to the number 13 through Western media (films, imported products) means that some Japanese individuals may hold mild triskaidekaphobic beliefs, though this is unlikely to approach the deep-seated cultural fear found in Western nations.¹⁰ Third, we analyzed raw accident counts rather than rates per vehicle-kilometer traveled, as daily traffic volume data were not available at the national level. However, because our primary comparisons are between Fridays (same day of week), systematic differences in traffic volume are minimized; the case-crossover design further controls for within-month temporal trends. The consistency of results across the covariate-adjusted panel model (which controls for holidays, weather, and seasonal patterns that affect traffic volume) provides additional reassurance, though residual confounding by unmeasured traffic volume cannot be fully excluded. Fourth, the study period includes the COVID-19 pandemic, which altered traffic patterns; however, sensitivity analyses excluding 2020 confirmed our findings. Fifth, the NPA dataset does not include driver sex, precluding direct replication of Näyhä's sex-stratified analysis. Sixth, the deviance-to-degrees-of-freedom ratios of 0.03 and 0.02 obtained from an alternative national-level generalized linear model (GLM) specification with α fixed at 1.0 should be read as suggesting possible overcorrection under that fixed- α assumption rather than as unqualified evidence of fit adequacy; the primary panel specification here estimates α by maximum likelihood ($\alpha_{NB2} \approx 0.023$), Pearson residuals have mean +0.000 and SD 1.01 with mild fat tails beyond $|r| > 3$ (Figure S1), and the quasi-Poisson comparison agrees to four decimals with a Pearson χ^2/df scale factor of 1.44. With a 47-cluster count below the number of covariates on the prefecture axis, the two-way cluster meat matrix is rank-deficient on that axis; the Cameron-Miller pair cluster bootstrap on the prefecture axis (500 iterations) was used as a sensitivity check and also failed to reject the null, though it does not resample date-clustered variance and therefore under-covers that component. Seventh, as an ecological study, we cannot make individual-level causal inferences; the absence of a population-level effect does not preclude the possibility that a small subset of individuals alter their behavior on Friday the 13th.

Conclusion

Friday the 13th was not associated with a detectable population-level increase in traffic accidents in Japan. When Western analytical methods are faithfully reproduced in a culture where 13 is not a traditionally salient unlucky number, no large effect emerges, although small effects cannot be excluded by our data. Although this pattern is consistent with a cultural-mediation hypothesis, the accumulated evidence—

including null results from recent large Western studies—more strongly supports a universal null effect. Regardless of the underlying explanation, the practical implication is the same: the roads of Japan are no more dangerous on Friday the 13th than on any other Friday. Scanlon’s advice to “stay at home” on Friday the 13th may be prudent counsel for the superstitious, but in Japan, there is no evidence-based reason to check the calendar before starting the car.

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Author Contributions (CRediT)

Mizuki Shirai: Conceptualization, Methodology, Investigation, Data Curation, Formal Analysis, Writing - Original Draft, Writing - Review & Editing, Visualization, Project Administration.

Conflict of Interest

The author declares no conflicts of interest per ICMJE guidelines.

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Data Availability

Traffic accident data are publicly available from the National Police Agency of Japan open data portal (https://www.npa.go.jp/publications/statistics/koutsuu/opendata/index_opendata.html). Weather data are from the Japan Meteorological Agency (<https://www.jma.go.jp/>). Analysis code is available at <https://github.com/rehabilitation-collaboration/friday13th>.

Tables

Table 1. Daily accident counts on Friday the 13th versus same-month other Fridays

Friday the 13th	Accidents	Same-month other Fridays (mean)	Ratio
2019-09-13	1,110	1,249	0.89
2019-12-13	1,343	1,263	1.06
2020-03-13	960	937	1.02
2020-11-13	1,178	1,052	1.12
2021-08-13	713	946	0.75
2022-05-13	1,022	905	1.13
2023-01-13	1,022	852	1.20
2023-10-13	1,081	1,075	1.01
2024-09-13	977	896	1.09
2024-12-13	863	733	1.18
Mean (arithmetic per-pair ratio)	1,026.9	990.9	1.05
Geometric-mean count ratio (case-crossover permutation)	—	—	1.04

“Same-month other Fridays” refers to the arithmetic mean of all Fridays in the same calendar month excluding the Friday the 13th itself (typically Fri 6, 20, and 27). This within-month control set differs from the “all other Fridays” comparison used in the Results text and Table 2, which pools all 303 non-13th Fridays across the study period, and from the Scanlon-method comparison (Friday 13th vs. same-month Friday 6th only). The bolded “Mean (arithmetic per-pair ratio)” row is the arithmetic average of the ten per-pair case-to-control ratios (1.05) — an alternative summary reported for reader transparency but not used as an inferential estimand. The “Geometric-mean count ratio” row (1.04) is $\exp(\text{mean of the ten log-ratios})$ and is the estimand of the sign-flip permutation and conditional-Poisson case-crossover tests reported in the Methods and Results. The two summaries are numerically close because the per-pair ratios sit near 1, but they answer slightly different statistical questions and are not interchangeable in downstream inference.

Table 2. Sensitivity analyses

Scenario	n(Fri 13th)	n(Other Fri)	Mean Fri 13th	Mean Other	RR (95% CI)	p-value
All data	10	303	1,026.9	988.6	1.04 (0.93-1.15)	0.50
Excl. COVID (2020)	8	253	1,016.4	992.0	1.02 (0.90-1.16)	0.70
Excl. Obon & New Year	9	295	1,061.8	997.9	1.06 (0.97-1.16)	0.26
Excl. holidays	10	283	1,026.9	1,004.7	1.02 (0.92-1.14)	0.67

Scenario	n(Fri 13th)	n(Other Fri)	Mean Fri 13th	Mean Other	RR (95% CI)	p-value
Excl. COVID + holidays	8	235	1,016.4	1,009.0	1.01 (0.89-1.14)	0.90

RR and p-values in this table are from independent two-sample t-tests comparing Friday the 13th against all other Fridays within each scenario, in contrast to the Scanlon paired t-test ($p = 0.47$) reported in the Results text.

Table 3. Cross-national comparison of Friday the 13th studies

Country	Study	Period	N	Outcome	Design	Effect (RR/OR)	Significant	Cultural fear of 13
UK	Scanlon 1993	multiple years	110	Hospital admissions	Paired	1.44	Yes	Strong
Finland	Näyhä 2002	1971-1997	41 (women)	Traffic fatalities	NegBin	Women 1.63 / Men 1.02	Women only	Moderate
Finland	Radun 2004	1989-2002	21 Fri 13ths	Injury accidents	Matched	~1.0	No	Moderate
Netherlands	CVS 2008	~2 years	7,500/day	Insurance claims	Descriptive	0.96	No	Moderate
Germany	Schuld 2011	2001-2010	27,914	Surgical outcomes	Retrospective	~1.0	No	Strong
USA	Lo 2012	8 years	49,094	ED visits	Multi-control	<1.0 (penetrating OR = 1.65)	Penetrating only	Strong
Canada	Ranganathan 2024	2007-2019	19,747	Surgical outcomes	Population cohort	1.02 (adjusted OR)	No	Strong
USA	Shekhar 2025	N/A	N/A	EMS volume	Letter	~1.0	No	Strong
Japan	Present study	2019-2024	1,884,793	Traffic accidents	Multiple	1.01-1.04	No	Absent

RR = rate ratio; OR = odds ratio; NegBin = negative binomial regression. N refers to total events analyzed except where noted (e.g., “21 Fri13s” = number of Friday the 13th dates, “41 (women)” = number of female traffic deaths). Effect estimates are rate ratios unless otherwise specified (e.g., Ranganathan 2024 and the Lo 2012 penetrating-trauma row are odds ratios, as noted in-cell). “N/A” indicates values not accessible from the original publication (Shekhar 2025 study, which was published as a letter without full descriptive statistics). The “Cultural fear of 13” classification is a qualitative

categorization based on prior cultural and ethnographic descriptions; it is intended to be interpreted descriptively rather than as a validated psychometric measure.

Table 4. Subgroup analyses: Friday the 13th vs. all other Fridays

Subgroup	Mean Fri13	Mean other Fri	Crude RR	Raw p (Welch)	Bonferroni p ($\times 8$)	NB-adjusted count ratio [95% CI]	NB p
Fatal accidents	9.90	7.57	1.307	0.087	0.696	1.227 [0.9997, 1.5061] †	0.050
Injury accidents	1,017.0	981.0	1.037	0.528	1.000	0.991 [0.914, 1.075]	0.833
Age ≤ 24 (young)	114.2	112.6	1.014	0.786	1.000	0.990 [0.905, 1.083]	0.820
Age 25-44 (mid-lower)	317.9	314.0	1.012	0.854	1.000	0.973 [0.891, 1.063]	0.543
Age 45-64 (mid-upper)	364.8	344.3	1.059	0.284	1.000	1.018 [0.937, 1.107]	0.674
Age ≥ 65 (elderly)	229.8	217.5	1.057	0.509	1.000	0.980 [0.888, 1.082]	0.694
Daytime	687.1	720.4	0.954	0.380	1.000	0.962 [0.884, 1.047]	0.366
Nighttime	339.8	268.2	1.267	0.142	1.000	1.065 [0.958, 1.183]	0.247

Family-wise Bonferroni threshold across the 8 pre-specified subgroups: $\alpha = 0.05 / 8 = 0.00625$. Raw p is the Welch (unequal-variance) two-sided t-test of Fri13 daily counts against all other Friday daily counts ($n = 10$ vs. 303). “NB-adjusted count ratio” is the exponentiated Friday-the-13th coefficient from a national-level negative binomial (NB2) regression with year, month, weekday, and holiday-flag controls; the dispersion parameter α was estimated jointly by maximum likelihood. † The fatal-accident NB2 maximum-likelihood optimizer (both BFGS and Nelder-Mead) failed to converge on the sparse Fri13 fatal counts (mean 9.9 accidents per Friday); the reported value is a Poisson approximation, which is expected to under-cover the true variance and should be interpreted as an approximate bracket rather than an exact confidence interval. None of the 8 subgroups survived Bonferroni correction.

Figures

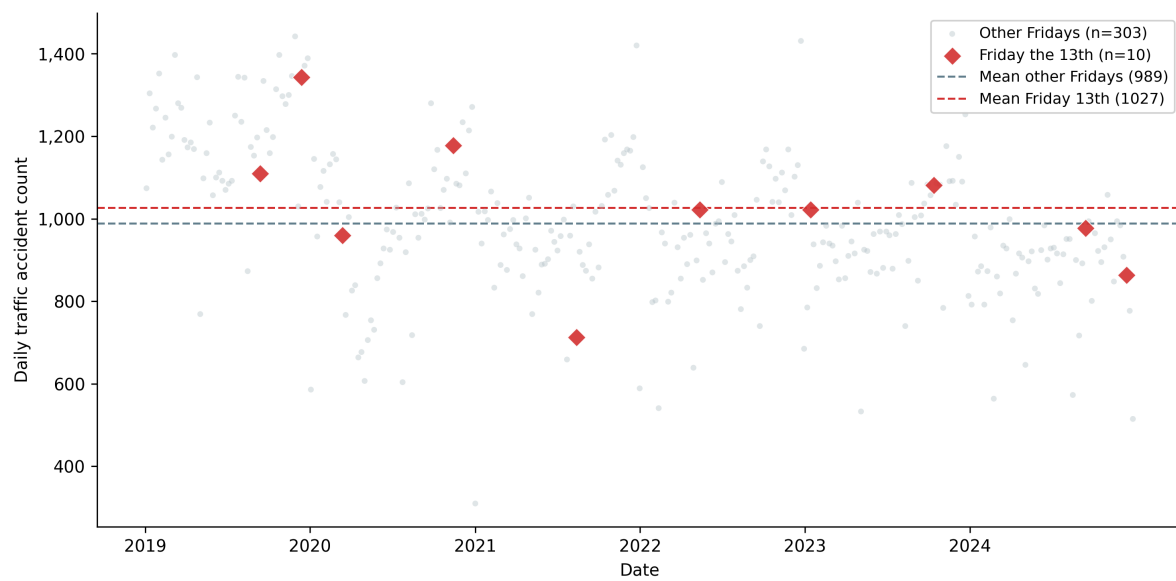


Figure 1. Daily traffic accident counts on Friday the 13th (red points) compared with all other Fridays (gray points) during 2019-2024 in Japan. The dashed horizontal lines indicate group means. No systematic elevation is observed for Friday the 13th dates.

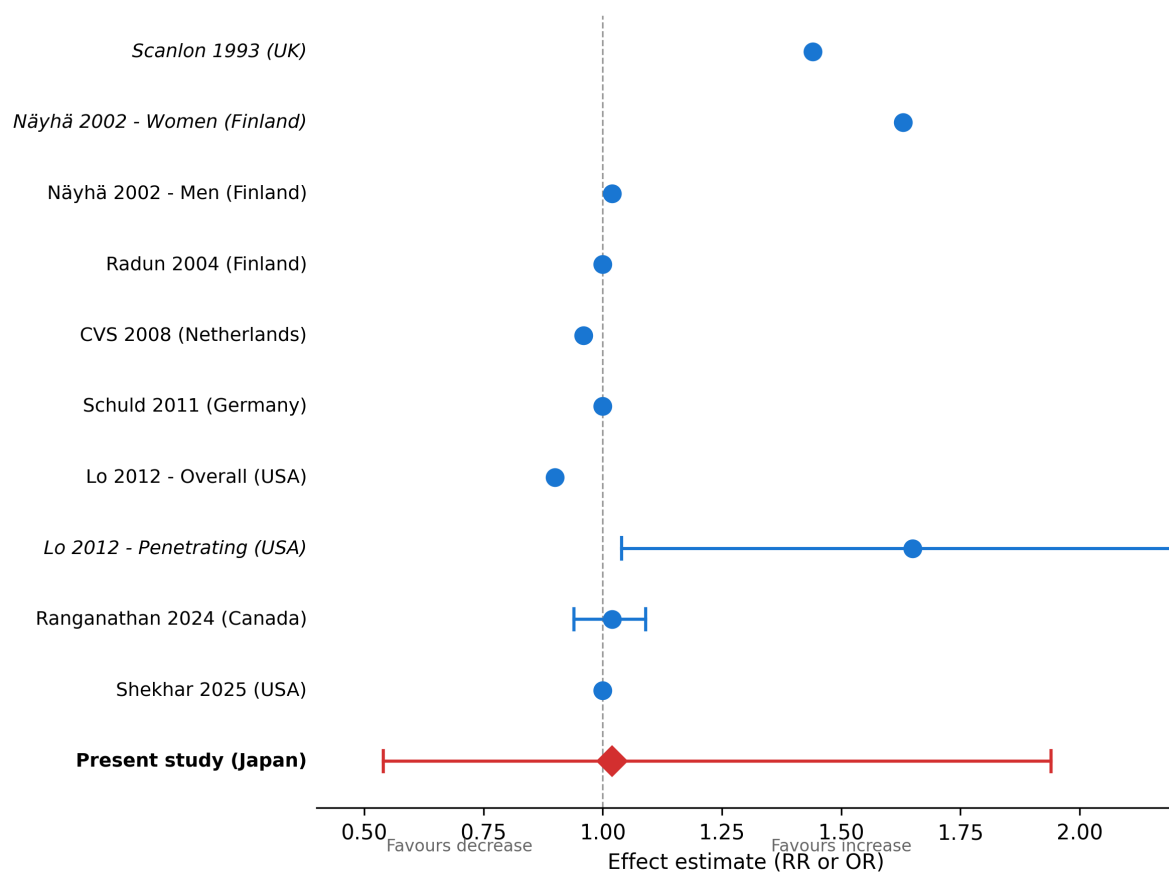


Figure 2. Cross-national forest plot of Friday the 13th effect estimates. Studies are ordered by publication year. Point estimates and 95% confidence intervals are shown where available. The vertical dashed line at $RR=1.0$ indicates no effect. Japan (present study) is highlighted with the largest sample size and a point estimate near unity.

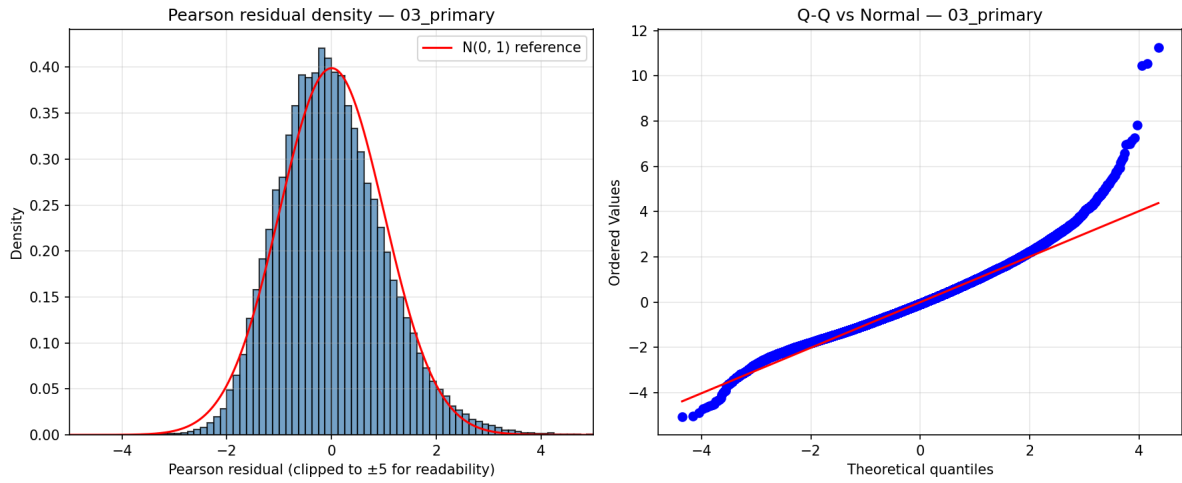


Figure S1. Pearson residual distribution for the primary weather-and-holiday-adjusted NB2 panel model ($n = 103,008$ prefecture-days). Empirical mean = $+0.000$, standard deviation = 1.01 ; 4.4% of residuals exceed $|r| > 2$ (versus 4.55% expected under a standard normal) and 0.69% exceed $|r| > 3$ (versus 0.27% expected). The mild fat tail beyond ± 3 is consistent with a small number of extreme-event days that count regression cannot fully absorb and does not indicate systematic model misspecification for the Friday-the-13th estimand.

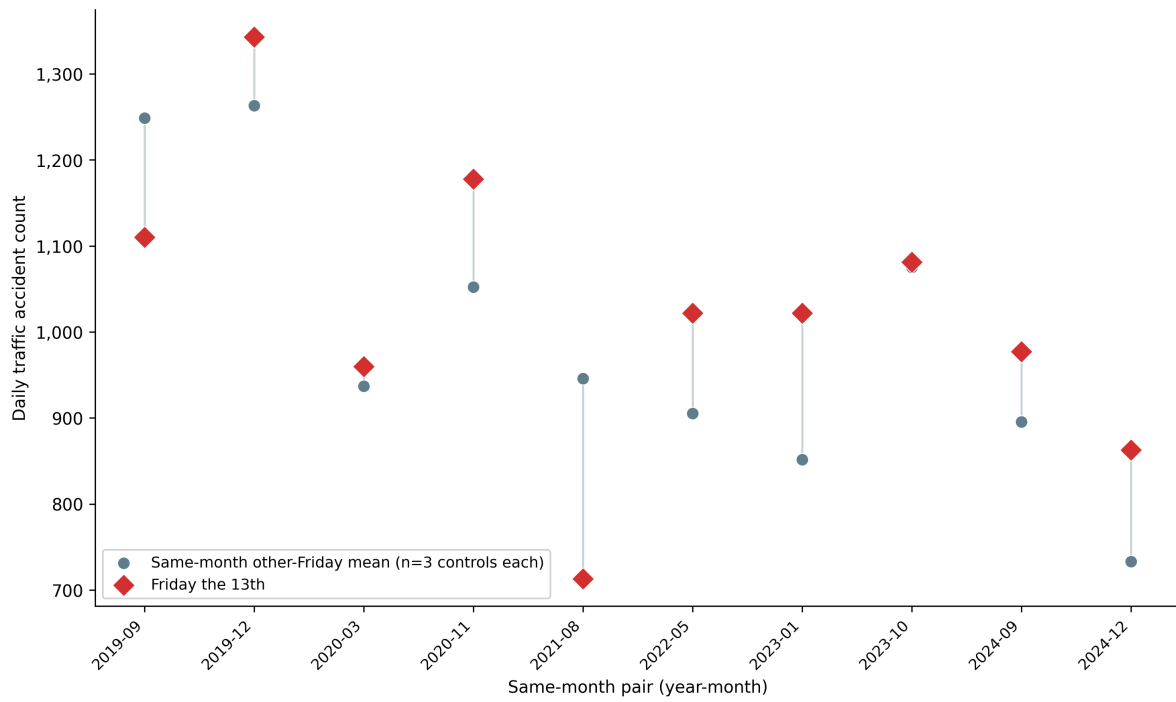


Figure S2. Same-month pair plot for the 10 Fri13-containing months of 2019-2024. Each connecting line represents one month (10 lines total); the left endpoint (red diamond) is the daily accident count on Friday the 13th and the right endpoint (blue circle) is the mean of the other Fridays in the same calendar month. Eight of ten pairs favor Fri13 > control (consistent with the exact sign test result reported in the Results, 8 of 10 pairs, $p = 0.11$), including a visible downward outlier in August 2021 (Obon period).

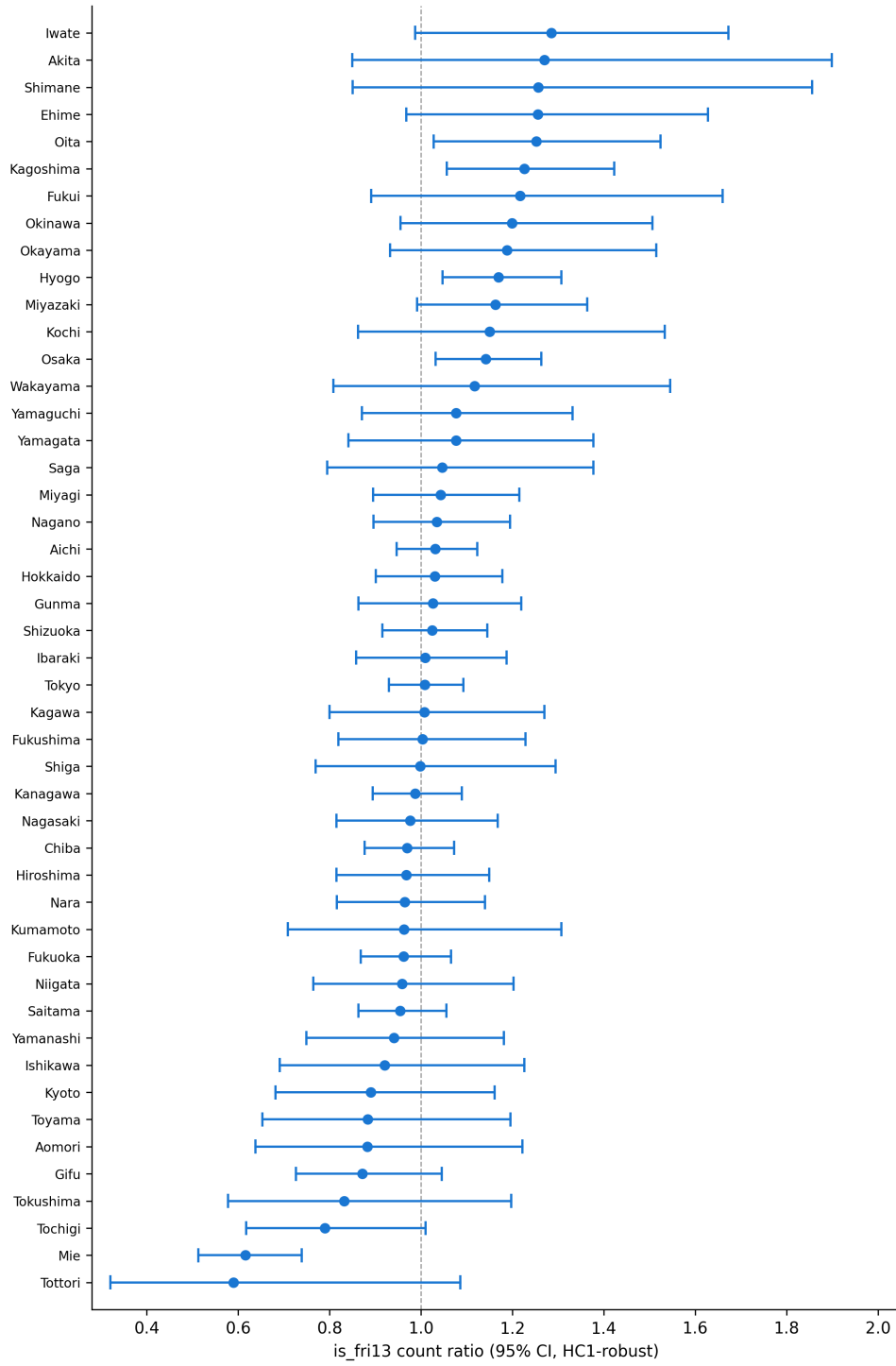


Figure S3. Forest plot of Friday-the-13th count ratios for each of the 47 prefectures, sorted by point estimate. Each row shows the prefecture-specific NB2 count ratio and its HC1 heteroscedasticity-robust 95% confidence interval; the vertical dashed line at 1.0 indicates no effect. Ratios range from 0.59 [0.32, 1.09] (Tottori) to 1.29 [0.99, 1.67] (Iwate), with a median of 1.01. Five prefectures had 95% CIs excluding 1.0 at the nominal $\alpha = 0.05$ level (11%, vs. 2.25 expected under the null; one-sided binomial excess-test $p = 0.08$). Under a family-wise Bonferroni correction only Mie prefecture remained significant; the Benjamini-Hochberg FDR procedure at $q = 0.05$ also retained only Mie. Ehime and Okayama required dropping the snow-depth covariate for those prefecture-specific fits (perfect collinearity with snowfall on their 1-2 snow days); all 47 prefectures converged. The overall distribution is centered

on 1.0 rather than showing a systematic Friday-the-13th shift, consistent with the primary panel and case-crossover null-everywhere results.